

```
In [1]: import numpy as np
import pandas as pd
import matplotlib as plt
import seaborn as sns
```

```
In [3]: df = pd.read_csv(r"C:\Users\Suresh Kunalala\Downloads\Inc_Exp_Data.csv")
df
```

Out[3]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income	Highest_Qualified_Member	No_of_E
0	5000	8000	3	2000	64200	Under-Graduate	
1	6000	7000	2	3000	79920	Illiterate	
2	10000	4500	2	0	112800	Under-Graduate	
3	10000	2000	1	0	97200	Illiterate	
4	12500	12000	2	3000	147000	Graduate	
5	14000	8000	2	0	196560	Graduate	
6	15000	16000	3	35000	167400	Post-Graduate	
7	18000	20000	5	8000	216000	Graduate	
8	19000	9000	2	0	218880	Under-Graduate	
9	20000	9000	4	0	220800	Under-Graduate	
10	20000	18000	4	8000	278400	Under-Graduate	
11	22000	25000	6	12000	279840	Illiterate	
12	23400	5000	3	0	292032	Illiterate	
13	24000	10500	6	0	316800	Graduate	
14	24000	10000	4	0	244800	Graduate	
15	25000	12300	3	0	246000	Graduate	
16	25000	20000	3	3500	261000	Graduate	
17	25000	10000	6	0	258000	Under-Graduate	
18	29000	6600	2	2000	348000	Graduate	
19	30000	13000	4	0	385200	Graduate	
20	30500	25000	5	5000	351360	Under-Graduate	
21	32000	15000	4	0	445440	Professional	

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income	Highest_Qualified_Member	No_of_E
22	34000	19000	6	0	330480	Professional	
23	34000	25000	3	4000	469200	Professional	
24	35000	12000	3	0	466200	Graduate	
25	35000	25000	4	0	449400	Professional	
26	39000	8000	4	0	556920	Under-Graduate	
27	40000	10000	4	0	412800	Under-Graduate	
28	42000	15000	4	0	488880	Graduate	
29	43000	12000	4	0	619200	Graduate	
30	45000	25000	6	0	523800	Graduate	
31	45000	40000	6	3500	507600	Professional	
32	45000	10000	2	1000	437400	Post-Graduate	
33	45000	22000	4	2500	610200	Post-Graduate	
34	46000	25000	5	3500	596160	Graduate	
35	47000	15000	7	0	456840	Professional	
36	50000	20000	4	0	570000	Professional	
37	50500	20000	3	0	581760	Professional	
38	55000	45000	6	12000	600600	Graduate	
39	60000	10000	3	0	590400	Post-Graduate	
40	60000	50000	6	10000	590400	Graduate	
41	65000	20000	4	5000	647400	Illiterate	
42	70000	9000	2	0	756000	Graduate	
43	80000	20000	4	0	1075200	Graduate	

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income	Highest_Qualified_Member	No_of_Ea
44	85000	25000	5	0	1142400	Under-Graduate	
45	90000	48000	7	0	885600	Post-Graduate	
46	98000	25000	5	0	1152480	Professional	
47	100000	30000	6	0	1404000	Graduate	
48	100000	50000	4	20000	1032000	Professional	
49	100000	40000	6	10000	1320000	Post-Graduate	

```
In [4]: df.head()
```

Out[4]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income	Highest_Qualified_Member	No_of_Ear
0	5000	8000	3	2000	64200	Under-Graduate	
1	6000	7000	2	3000	79920	Illiterate	
2	10000	4500	2	0	112800	Under-Graduate	
3	10000	2000	1	0	97200	Illiterate	
4	12500	12000	2	3000	147000	Graduate	

```
In [5]: len(df)
```

Out[5]: 50

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Mthly_HH_Income        50 non-null    int64
1   Mthly_HH_Expense       50 non-null    int64
2   No_of_Fly_Members      50 non-null    int64
3   Emi_or_Rent_Amt        50 non-null    int64
4   Annual_HH_Income       50 non-null    int64
5   Highest_Qualified_Member 50 non-null    object
6   No_of_Earning_Members  50 non-null    int64
dtypes: int64(6), object(1)
memory usage: 2.9+ KB
```

```
In [8]: df.shape
```

Out[8]: (50, 7)

```
In [12]: df.describe().T
```

Out[12]:

	count	mean	std	min	25%	50%	75%	max
Mthly_HH_Income	50.0	41558.00	26097.908979	5000.0	23550.0	35000.0	50375.0	100000.0
Mthly_HH_Expense	50.0	18818.00	12090.216824	2000.0	10000.0	15500.0	25000.0	50000.0
No_of_Fly_Members	50.0	4.06	1.517382	1.0	3.0	4.0	5.0	7.0
Emi_or_Rent_Amt	50.0	3060.00	6241.434948	0.0	0.0	0.0	3500.0	35000.0
Annual_HH_Income	50.0	490019.04	320135.792123	64200.0	258750.0	447420.0	594720.0	1404000.0
No_of_Earning_Members	50.0	1.46	0.734291	1.0	1.0	1.0	2.0	4.0

```
In [14]: df.isna()
```

Out[14]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income	Highest_Qualified_Member	No_of_E
0	False	False	False	False	False	False	
1	False	False	False	False	False	False	
2	False	False	False	False	False	False	
3	False	False	False	False	False	False	
4	False	False	False	False	False	False	
5	False	False	False	False	False	False	
6	False	False	False	False	False	False	
7	False	False	False	False	False	False	
8	False	False	False	False	False	False	
9	False	False	False	False	False	False	
10	False	False	False	False	False	False	
11	False	False	False	False	False	False	
12	False	False	False	False	False	False	
13	False	False	False	False	False	False	
14	False	False	False	False	False	False	
15	False	False	False	False	False	False	
16	False	False	False	False	False	False	
17	False	False	False	False	False	False	
18	False	False	False	False	False	False	
19	False	False	False	False	False	False	
20	False	False	False	False	False	False	
21	False	False	False	False	False	False	

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income	Highest_Qualified_Member	No_of_E
22	False	False	False	False	False	False	
23	False	False	False	False	False	False	
24	False	False	False	False	False	False	
25	False	False	False	False	False	False	
26	False	False	False	False	False	False	
27	False	False	False	False	False	False	
28	False	False	False	False	False	False	
29	False	False	False	False	False	False	
30	False	False	False	False	False	False	
31	False	False	False	False	False	False	
32	False	False	False	False	False	False	
33	False	False	False	False	False	False	
34	False	False	False	False	False	False	
35	False	False	False	False	False	False	
36	False	False	False	False	False	False	
37	False	False	False	False	False	False	
38	False	False	False	False	False	False	
39	False	False	False	False	False	False	
40	False	False	False	False	False	False	
41	False	False	False	False	False	False	
42	False	False	False	False	False	False	
43	False	False	False	False	False	False	

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income	Highest_Qualified_Member	No_of_Ea
44	False	False	False	False	False	False	
45	False	False	False	False	False	False	
46	False	False	False	False	False	False	
47	False	False	False	False	False	False	
48	False	False	False	False	False	False	
49	False	False	False	False	False	False	

```
In [15]: df.isna().any()
```

```
Out[15]: Mthly_HH_Income      False
Mthly_HH_Expense      False
No_of_Fly_Members      False
Emi_or_Rent_Amt        False
Annual_HH_Income      False
Highest_Qualified_Member False
No_of_Earning_Members  False
dtype: bool
```

```
In [16]: df.columns
```

```
Out[16]: Index(['Mthly_HH_Income', 'Mthly_HH_Expense', 'No_of_Fly_Members',
               'Emi_or_Rent_Amt', 'Annual_HH_Income', 'Highest_Qualified_Member',
               'No_of_Earning_Members'],
              dtype='object')
```

```
In [18]: print(df['Mthly_HH_Expense'].mean())
```

```
18818.0
```

```
In [19]: print(df['Mthly_HH_Expense'].median())
```

```
15500.0
```

```
In [20]: print(df['Mthly_HH_Expense'].mode())
```

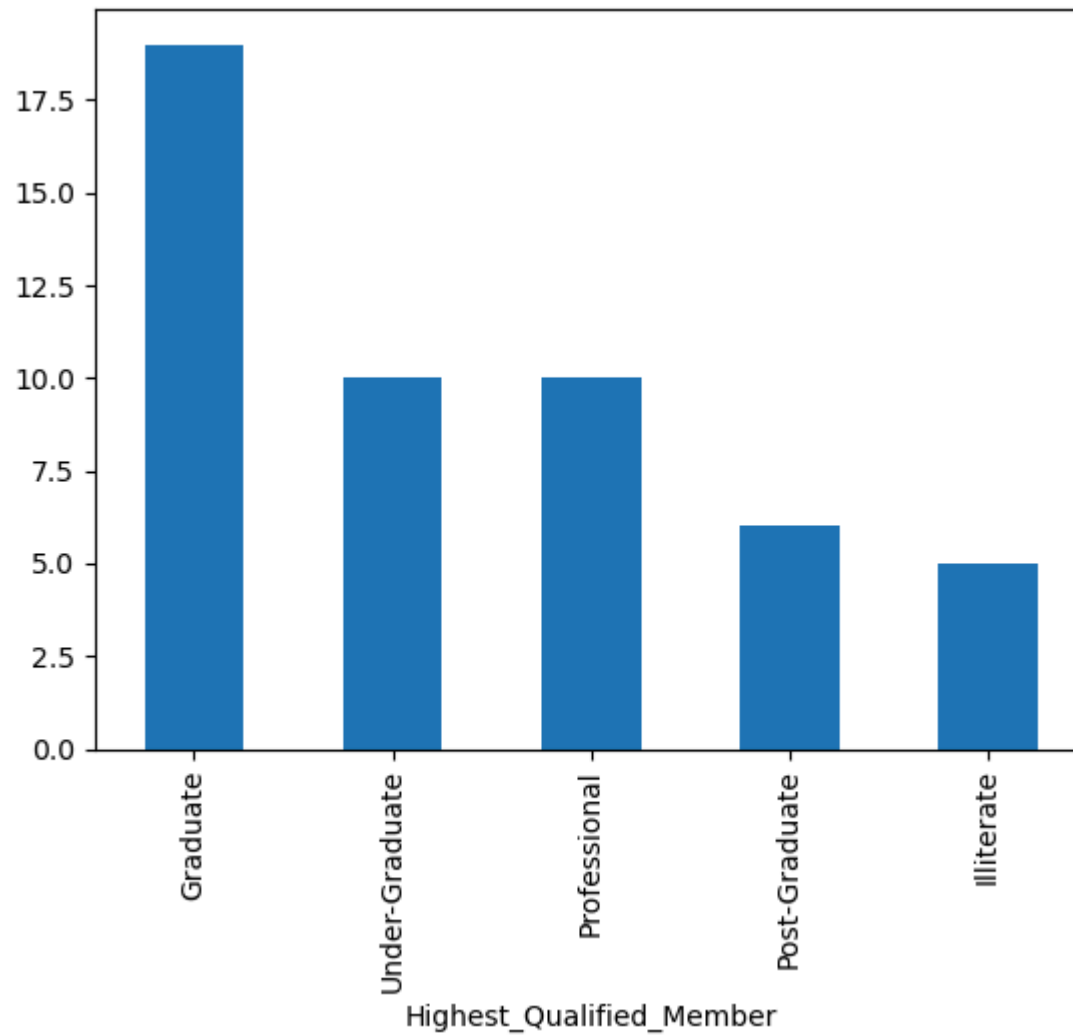
```
0    25000
```

```
Name: Mthly_HH_Expense, dtype: int64
```

```
In [ ]: mth_exp_tmp = pd.crosstab(index=df['Mthly_HH_Expense'], columns = 'count')  
mth_exp_tmp.reset_index(inplace = True)  
mth_exp_tmp[mth_exp_tmp['count'] == df.Mthly_HH_Expense.value_counts]
```

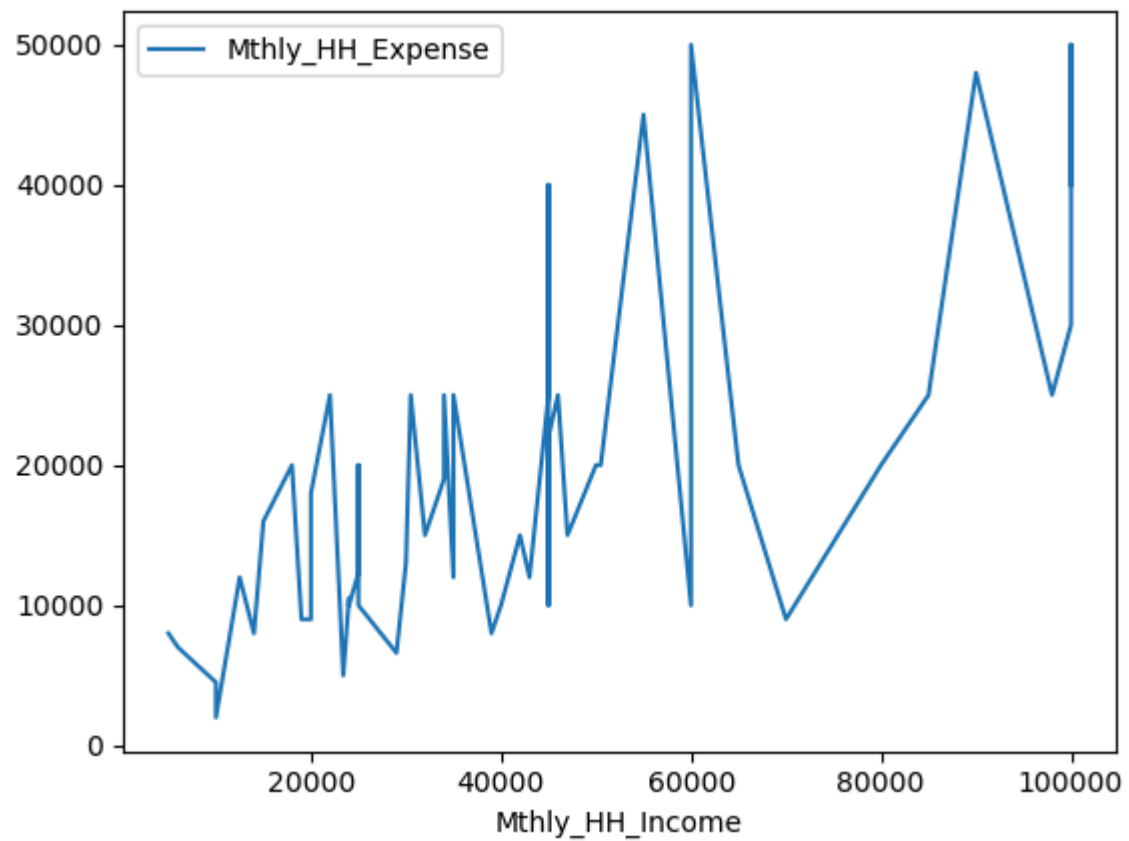
```
In [22]: df['Highest_Qualified_Member'].value_counts().plot(kind = 'bar')
```

```
Out[22]: <Axes: xlabel='Highest_Qualified_Member'>
```



```
In [25]: df.plot(x = 'Mthly_HH_Income', y = 'Mthly_HH_Expense')
```

```
Out[25]: <Axes: xlabel='Mthly_HH_Income'>
```



```
In [28]: df['Mthly_HH_Expense'].var()
```

```
Out[28]: 146173342.85714287
```

```
In [29]: df['Mthly_HH_Expense'].std()
```

```
Out[29]: 12090.216824240286
```

```
In [30]: df
```

Out[30]:

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income	Highest_Qualified_Member	No_of_E
0	5000	8000	3	2000	64200	Under-Graduate	
1	6000	7000	2	3000	79920	Illiterate	
2	10000	4500	2	0	112800	Under-Graduate	
3	10000	2000	1	0	97200	Illiterate	
4	12500	12000	2	3000	147000	Graduate	
5	14000	8000	2	0	196560	Graduate	
6	15000	16000	3	35000	167400	Post-Graduate	
7	18000	20000	5	8000	216000	Graduate	
8	19000	9000	2	0	218880	Under-Graduate	
9	20000	9000	4	0	220800	Under-Graduate	
10	20000	18000	4	8000	278400	Under-Graduate	
11	22000	25000	6	12000	279840	Illiterate	
12	23400	5000	3	0	292032	Illiterate	
13	24000	10500	6	0	316800	Graduate	
14	24000	10000	4	0	244800	Graduate	
15	25000	12300	3	0	246000	Graduate	
16	25000	20000	3	3500	261000	Graduate	
17	25000	10000	6	0	258000	Under-Graduate	
18	29000	6600	2	2000	348000	Graduate	
19	30000	13000	4	0	385200	Graduate	
20	30500	25000	5	5000	351360	Under-Graduate	
21	32000	15000	4	0	445440	Professional	

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income	Highest_Qualified_Member	No_of_E
22	34000	19000	6	0	330480	Professional	
23	34000	25000	3	4000	469200	Professional	
24	35000	12000	3	0	466200	Graduate	
25	35000	25000	4	0	449400	Professional	
26	39000	8000	4	0	556920	Under-Graduate	
27	40000	10000	4	0	412800	Under-Graduate	
28	42000	15000	4	0	488880	Graduate	
29	43000	12000	4	0	619200	Graduate	
30	45000	25000	6	0	523800	Graduate	
31	45000	40000	6	3500	507600	Professional	
32	45000	10000	2	1000	437400	Post-Graduate	
33	45000	22000	4	2500	610200	Post-Graduate	
34	46000	25000	5	3500	596160	Graduate	
35	47000	15000	7	0	456840	Professional	
36	50000	20000	4	0	570000	Professional	
37	50500	20000	3	0	581760	Professional	
38	55000	45000	6	12000	600600	Graduate	
39	60000	10000	3	0	590400	Post-Graduate	
40	60000	50000	6	10000	590400	Graduate	
41	65000	20000	4	5000	647400	Illiterate	
42	70000	9000	2	0	756000	Graduate	
43	80000	20000	4	0	1075200	Graduate	

	Mthly_HH_Income	Mthly_HH_Expense	No_of_Fly_Members	Emi_or_Rent_Amt	Annual_HH_Income	Highest_Qualified_Member	No_of_Ei
44	85000	25000	5	0	1142400	Under-Graduate	
45	90000	48000	7	0	885600	Post-Graduate	
46	98000	25000	5	0	1152480	Professional	
47	100000	30000	6	0	1404000	Graduate	
48	100000	50000	4	20000	1032000	Professional	
49	100000	40000	6	10000	1320000	Post-Graduate	

```
In [35]: print(df['Mthly_HH_Expense'].kurtosis())
```

0.9424897886350738

```
In [ ]:
```

```
In [ ]:
```

```
In [ ]:
```